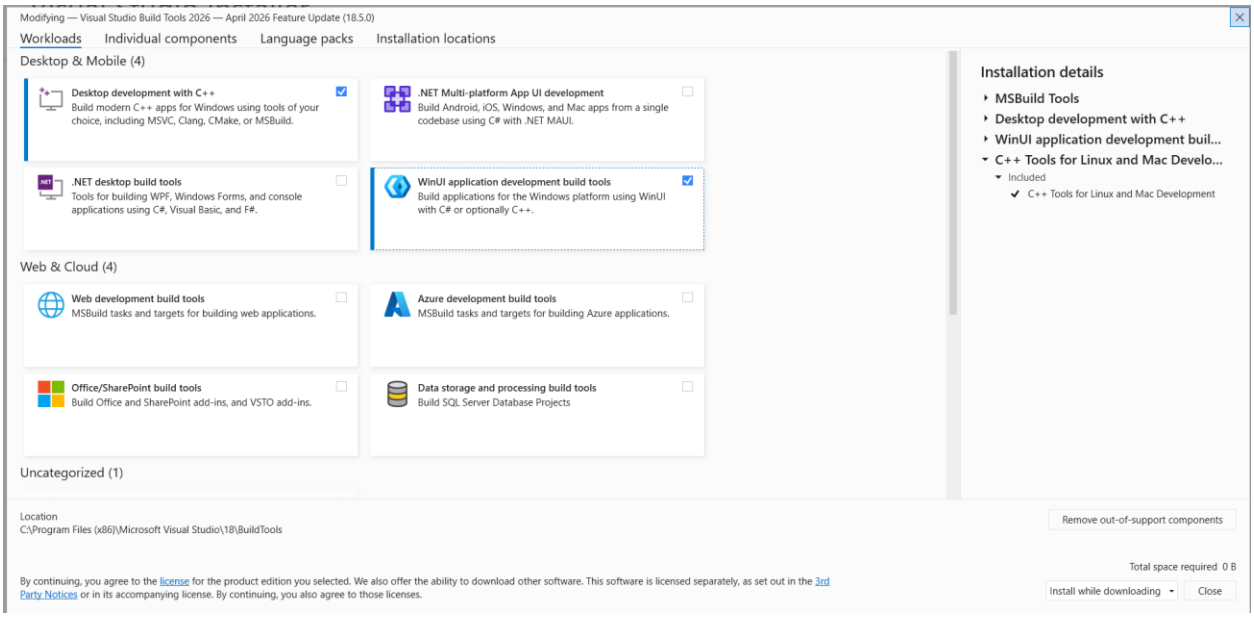
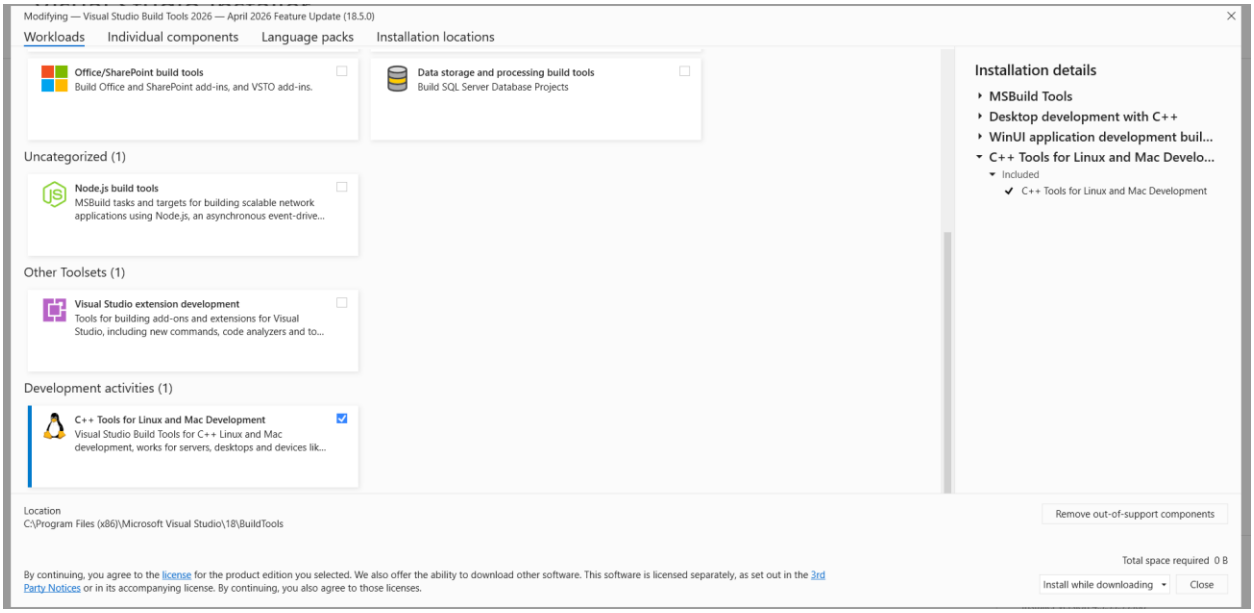
1. First install python 3.11.0 from https://www.python.org/downloads/release/python-3110/. When you are installing make sure to click on" Add to Path" option as well
2. Second Install Visual CPP tools from

   https://visualstudio.microsoft.com/visual-cpp-build-tools/

3. When you are installing the visual CPP tools initially It will download and install visual studio, after that you must click the respective packages in the below mentioned Screenshot.





4. Once you select, then click on install, please make sure your system should have 35GB free storage. It will download the packages and then install the downloaded

packages. It will take 30 minutes of time based on your internet speed and System Compatibility.

5. Now clone the git repository using command
6. **“git clone https://github.com/VenkataSaichandNeeli/Video-Surveillance-UCF-Crime-MOT-17.git”**
7. After that go to that directory
8. “cd Video Surveillance
9. Now Create and activate a virtual environment:
10. **python -m venv .venv**
11. **source .venv/bin/activate** # Linux / macOS
12. **.venv\Scripts\activate.bat** # Windows
13. Now after activation you will see (venv) at the begining of the terminal path.
14. Now go to https://github.com/cisco/openh264/releases/tag/v1.8.0 and download the openh264-1.8.0-win64.dll. Since i am using windows i download this. Based on your architecture and OS, download the appropriate asset. After that, Extract the zip and you can find openh264-1.8.0-win64.dll file. Now copy and paste the file exactly in your project root folder where you are running the run.py
15. Now we must install Pytorch from the link below. Make sure the version should be 2.6.0, not older than 2.6.0.
16. **pip install torch==2.6.0 torchvision==0.21.0 torchaudio==2.6.0 --index-url https://download.pytorch.org/whl/cu124 --upgrade**
17. Now Install the other packages from our requirements.txt using the below command
18. **Pip install –r requirements.txt**
19. Once you have installed the packages now, we must run the pipeline. To run the pipeline, you should place the UCF Crime Videos or MOT folders in the data folder, otherwise an absolute path should be given. A sample was already there.
20. To run the UCF crime clips, use the following command
21. **python run.py --video data/test_clip.mp4 --zones config/zones_template.json --output results/ucfcrime1 --preview --edit-zone**
22. Here I placed the UCF Crimes video in data folder and the name is test_clip, you can choose whatever the name you wanted to store the results here i have given results/ucfcrime1 i.e. the output video, the event logs, frame information will be stored in results/ucfcrime1 folder, after that if you wanted to enable the live preview use –preview otherwise ignore this, also if you wanted to manually draw a zone use –enable-zone otherwise ignore this as well. The zone editor options you can find in the terminal. Once you run the above command, initially it will preview the video, and you have the ability to draw the zone.

23. Now if you want to process a set of videos of UCF crime in a folder, use run_ucf.sh if you are using Linux or if you are using windows, use run_ucf.bat. But the structure should be data/ucf/Fighting/ Inside the Fighting Folder you need to place the individual .mp4 files.
24. Now if we want to run this pipeline for MOT17 data, use the below command.
25. **python run.py --video data/mot_test_clip_1/ --zones config/zones_template.json --output results/mot_result/ --mode sequence --evaluate --device cuda --preview**
26. Here inside the data folder, the format is like below
27. data/MOT17/

```
 MOT17-05-FRCNN/
   seqinfo.ini          Frame rate, resolution, sequence length
   img1/
  000001.jpg
    000002.jpg
    ...
   det/
    det.txt            Pre-computed FRCNN detections
   gt/
    Gt.txt
```

28. Sample files are in the data folder. Now if you wanted to run it for multiple clips of MOT17 use run_mot17.bat if you are using windows, if you are using linux use run_mot17_linux.sh. But the Directory structure should be data/MOT17/ inside this folder you will have sub folders mentioned as MOT17-05-FRCNN which are having of seqinfo, gt folder, det folder, img1 folders.
29. In the results folder mot17 folder you can find the event logs, and output video of every folder which we processed.